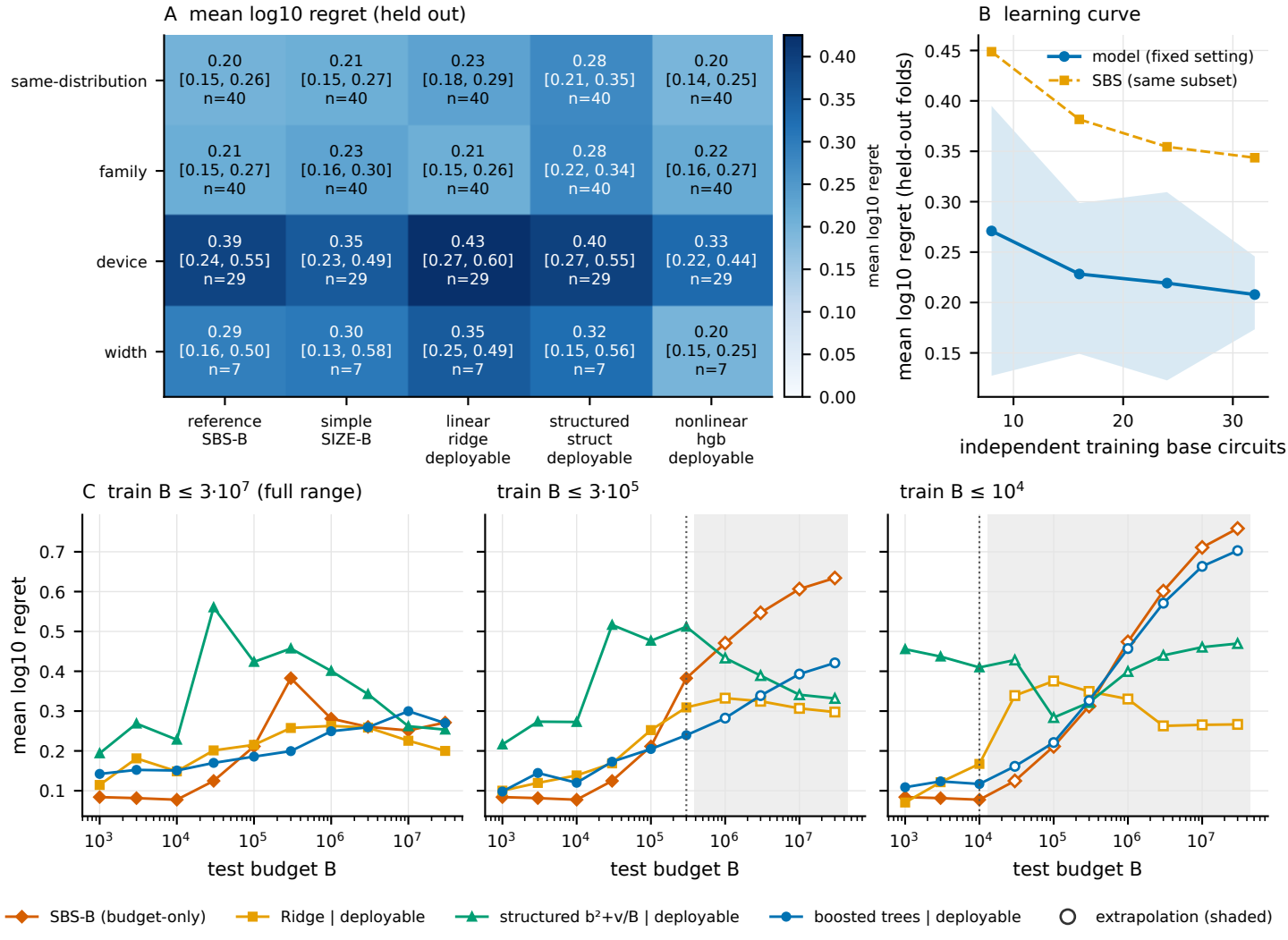
Simulated observations (emulated fake-device trials); no hardware data.
A mean log10 regret (held out)
same-distribution
family
device
width
0.20 [0.15, 0.26] n=40
0.21 [0.15, 0.27] n=40
0.23 [0.18, 0.29] n=40
0.28 [0.21, 0.35] n=40
0.20 [0.14, 0.25] n=40
0.21 [0.15, 0.27] n=40
0.23 [0.16, 0.30] n=40
0.21 [0.15, 0.26] n=40
0.28 [0.22, 0.34] n=40
0.22 [0.16, 0.27] n=40
0.39 [0.24, 0.55] n=29
0.35 [0.23, 0.49] n=29
0.43 [0.27, 0.60] n=29
0.40 [0.27, 0.55] n=29
0.33 [0.22, 0.44] n=29
0.29 [0.16, 0.50] n=7
0.30 [0.13, 0.58] n=7
0.35 [0.25, 0.49] n=7
0.32 [0.15, 0.56] n=7
0.20 [0.15, 0.25] n=7
reference SBS-B
simple SIZE-B
linear ridge deployable
structured struct deployable
nonlinear hgb deployable
mean log10 regret
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
B learning curve
model (fixed setting)
SBS (same subset)
mean log10 regret (held-out folds)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
10
20
30
independent training base circuits
C train B ≤ 3·10^7 (full range)
train B ≤ 3·10^5
train B ≤ 10^4
mean log10 regret
0.7
0.6
0.5
0.4
0.3
0.2
0.1
10^3
10^4
10^5
10^6
10^7
test budget B
SBS-B (budget-only)
Ridge | deployable
structured b²+v/B | deployable
boosted trees | deployable
extrapolation (shaded)